CSE STUDY HUB

Generative AI & Prompt Engineering

UNIT 1 · Foundations of Generative AI and LLMs

What this unit is about

This unit builds understanding of **Foundations of Generative AI and LLMs**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. AI, machine learning and deep learning: relationship and differences

2. Generative vs discriminative models

3. Generative AI modalities: text, image, audio, video and code

4. Large Language Models (LLMs): tokens, vocabulary and next-token prediction

5. Training pipeline: data, pre-training, fine-tuning and alignment

6. Transformer basics: attention and context window

7. Model output, sampling and inference

8. Strengths and limits: hallucinations, bias, privacy, copyright and evaluation

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 1 NOTES · PART 1

AI, machine learning and deep learning: relationship and differences

Compare the alternatives in AI, machine learning and deep learning: relationship and differences using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Generative vs discriminative models

Compare the alternatives in Generative vs discriminative models using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Generative AI modalities: text, image, audio, video and code

Generative AI learns patterns from training data and produces new content such as text, images, audio or code. A language model predicts token sequences; plausible output is not proof of factual correctness.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Foundations of Generative AI and LLMs, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 2

Large Language Models (LLMs): tokens, vocabulary and next-token prediction

An LLM represents text as tokens and uses a neural architecture (commonly a Transformer) to predict likely next tokens conditioned on context. Context windows limit how much input it can consider at once.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Training pipeline: data, pre-training, fine-tuning and alignment

Training pipeline: data, pre-training, fine-tuning and alignment is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Transformer basics: attention and context window

Start with a precise definition of Transformer basics: attention and context window. Explain its purpose, the problem it solves, its main components and where it is used. In an exam answer, add one practical example and distinguish it from the nearest related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Foundations of Generative AI and LLMs, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 1 NOTES · PART 3

Model output, sampling and inference

Treat Model output, sampling and inference as a structured model. Name its components in order, explain the responsibility of each component, and show how information moves between them. Include a labelled block diagram and one use case.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Strengths and limits: hallucinations, bias, privacy, copyright and evaluation

For Strengths and limits: hallucinations, bias, privacy, copyright and evaluation, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Foundations of Generative AI and LLMs, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **AI, machine learning and deep learning: relationship and differences** — know the definition, core working and one example.

• **Generative vs discriminative models** — know the definition, core working and one example.

• **Generative AI modalities: text, image, audio, video and code** — know the definition, core working and one example.

• **Large Language Models (LLMs): tokens, vocabulary and next-token prediction** — know the definition, core working and one example.

• **Training pipeline: data, pre-training, fine-tuning and alignment** — know the definition, core working and one example.

• **Transformer basics: attention and context window** — know the definition, core working and one example.

• **Model output, sampling and inference** — know the definition, core working and one example.

• **Strengths and limits: hallucinations, bias, privacy, copyright and evaluation** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Foundations of Generative AI and LLMs. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define AI, machine learning and deep learning: relationship and differences and explain its importance.

Q2. Explain the working/steps involved in Generative vs discriminative models.

Q3. Differentiate between two related concepts from Foundations of Generative AI and LLMs.

Q4. Give one practical application of Strengths and limits: hallucinations, bias, privacy, copyright and evaluation and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Foundations of Generative AI and LLMs.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.